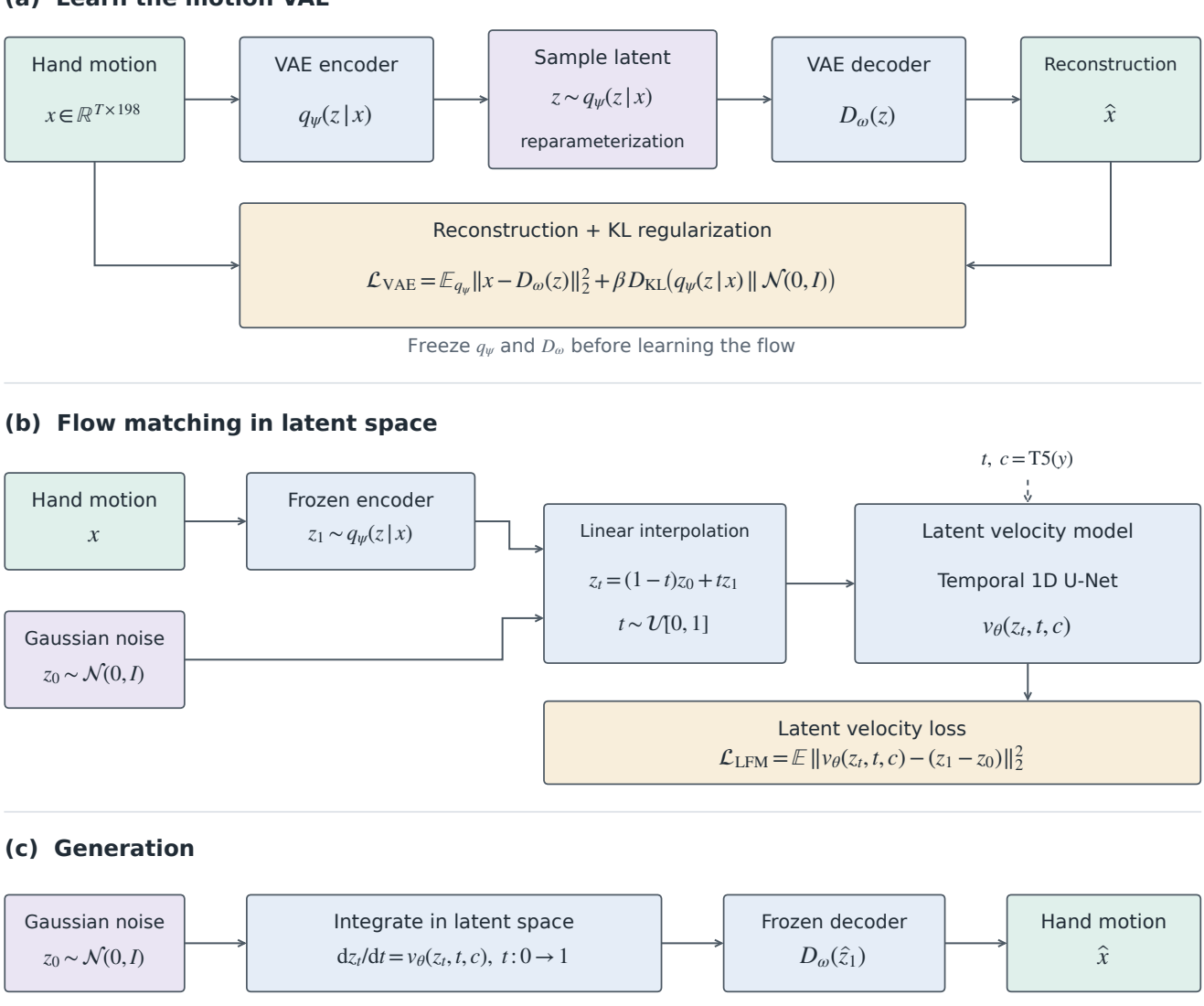
(a) Learn the motion VAE
Hand motion
$x \in \mathbb{R}^{T\times 198}$
VAE encoder
$q_\psi(z\,|\,x)$
Sample latent
$z \sim q_\psi(z\,|\,x)$
reparameterization
VAE decoder
$D_\omega(z)$
Reconstruction
$\hat{x}$
Reconstruction + KL regularization
$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\psi}\|x - D_\omega(z)\|_2^2 + \beta D_{\text{KL}}(q_\psi(z\,|\,x)\,\|\,\mathcal{N}(0,I))$
Freeze q_ψ and D_ω before learning the flow
(b) Flow matching in latent space
$t,\ c = \text{T5}(y)$
Hand motion
x
Frozen encoder
$z_1 \sim q_\psi(z\,|\,x)$
Gaussian noise
$z_0 \sim \mathcal{N}(0,I)$
Linear interpolation
$z_t = (1-t)z_0 + tz_1$
$t \sim \mathcal{U}[0,1]$
Latent velocity model
Temporal 1D U-Net
$v_\theta(z_t, t, c)$
Latent velocity loss
$\mathcal{L}_{\text{LFM}} = \mathbb{E}\,\|v_\theta(z_t,t,c) - (z_1 - z_0)\|_2^2$
(c) Generation
Gaussian noise
$z_0 \sim \mathcal{N}(0,I)$
Integrate in latent space
$\mathrm{d}z_t/\mathrm{d}t = v_\theta(z_t,t,c),\ t: 0 \to 1$
Frozen decoder
$D_\omega(\hat{z}_1)$
Hand motion
$\hat{x}$